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PRACTICAL 1

Task 1: Command execution using `fork()`, `exec()` and `wait()`

```
root@HP: ~/shellforge
GNU nano 7.2 week1_task1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%99s", command);

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child PID: %d\n", getpid());
        printf("Executing command: %s\n", command);

        execlp(command, command, (char *)NULL);

        perror("exec failed");
        return 1;
    }
    else
    {
        printf("Parent PID: %d\n", getpid());
        printf("Waiting for child process...\n");

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line

root@HP:~# cd ~/shellforge
root@HP:~/shellforge# nano week1_task1.c
root@HP:~/shellforge# gcc week1_task1.c -o week1_task1
root@HP:~/shellforge# ./week1_task1
Enter a Linux command: ls
Parent PID: 619
Waiting for child process...
Child PID: 620
Executing command: ls
session1.c.save week1_task1 week1_task1.c
Child process completed.
root@HP:~/shellforge#
```

Task 2: Using Linux terminal commands such as `uname`, `lscpu`, `lsblk`, `ps`, and `top`, investigate the relationship between hardware resources and operating system services.

```
root@HP:~/shellforge# uname -a
Linux HP 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux

root@HP:~/shellforge# lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                       8
  On-line CPU(s) list:       0-7
Vendor ID:                   GenuineIntel
Model name:                  Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz
CPU family:                  6
Model:                       142
Thread(s) per core:          2
Core(s) per socket:          4
Socket(s):                   1
Stepping:                    12
BogoMIPS:                    4224.01
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge m
ca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht sysc
all nx pdpe1gb rdtscp lm arch_perfmon rep_good nopl xt
opology cpuid tsc_known_freq pni pclmulqdq vmx ssse3 f
ma cx16 pdcm pcid sse4_1 sse4_2 movbe popcnt aes xsave
avx fl6c rdrand hypervisor lahf_lm abm 3dnowprefetch
ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid
ept_ad fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdse
ed adx smap clflushopt xsaveopt xsavec xgetbv1 xsaves
vnm1 md_clear flush_l1d arch_capabilities

Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:          Microsoft
  Virtualization type:        full
Caches (sum of all):
  L1d:                       128 KiB (4 instances)
  L1i:                       128 KiB (4 instances)
  L2:                         1 MiB (4 instances)
  L3:                         6 MiB (1 instance)
NUMA:
  NUMA node(s):               1
  NUMA node0 CPU(s):          0-7
Vulnerabilities:
  Gather data sampling:        Not affected
  Ghostwrite:                 Not affected
  Indirect target selection:    Mitigation; Aligned branch/return thunks
  Itlb multihit:              KVM: Mitigation: Split huge pages
  L1tf:                       Not affected
  Mds:                        Not affected
```

```

Ghostwrite: Not affected
Indirect target selection: Mitigation; Aligned branch/return thunks
Itlb multihit: KVM: Mitigation: Split huge pages
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Mitigation; Clear CPU buffers; SMT Host state unknown
Old microcode: Not affected
Reg file data sampling: Not affected
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI SW loop, KVM SW loop
Srbds: Unknown: Dependent on hypervisor status
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
root@HP:~/shellforge# |

root@HP:~/shellforge# lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
/
root@HP:~/shellforge# |

```

```

root@HP:~/shellforge# ps
  PID TTY          TIME CMD
  592 pts/0    00:00:00 bash
  631 pts/0    00:00:00 ps
root@HP:~/shellforge# ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.5  0.1  21740 13040 ?        Ss   04:48   0:01 /sbin/init
root         2  0.0  0.0   3180  2200 hvc0     Sl+  04:48   0:00 /init
root         7  0.0  0.0   3196  2132 hvc0     Sl+  04:48   0:00 plan9 --control-
root        59  0.1  0.1  34056 12560 ?        S<s  04:48   0:00 /usr/lib/systemd
root       107  0.1  0.0  25264  6552 ?        Ss   04:48   0:00 /usr/lib/systemd
systemd+   118  0.0  0.1  21344 13232 ?        Ss   04:48   0:00 /usr/lib/systemd
systemd+   119  0.0  0.0  91036  7996 ?        Ssl  04:48   0:00 /usr/lib/systemd
root       174  0.0  0.0   4244  2680 ?        Ss   04:48   0:00 /usr/sbin/cron -
message+   175  0.0  0.0   9600  5360 ?        Ss   04:48   0:00 @dbus-daemon --s
root       195  0.0  0.1  17984  8888 ?        Ss   04:48   0:00 /usr/lib/systemd
root       204  0.1  0.1 1756104 12996 ?        Ssl  04:48   0:00 /usr/libexec/wsl
root       228  0.0  0.0   3124  1976 tty1     Ss+  04:48   0:00 /sbin/agetty -o
syslog     232  0.0  0.0  222516  6036 ?        Ssl  04:48   0:00 /usr/sbin/rsyslo
root       282  0.1  0.2 107032 23136 ?        Ssl  04:48   0:00 /usr/bin/python3
root       367  0.0  0.0   6672  4496 pts/1    Ss   04:48   0:00 /bin/login -f
root       423  0.0  0.1  20320 11472 ?        Ss   04:48   0:00 /usr/lib/systemd
root       424  0.0  0.0   21168  3504 ?        S    04:48   0:00 (sd-pam)
root       450  0.0  0.0   6080  5324 pts/1    S+   04:48   0:00 -bash
root       587  0.0  0.0   3184  1112 ?        Ss   04:49   0:00 /init
root       588  0.0  0.0   3200  1252 ?        S    04:49   0:00 /init
root       592  0.0  0.0   6080  5368 pts/0    Ss   04:49   0:00 -bash
root       632  0.0  0.0   8288  4308 pts/0    R+   04:53   0:00 ps aux
root@HP:~/shellforge#

```

```

root@HP: ~/shellforge
top - 04:54:28 up 5 min, 1 user, load average: 0.11, 0.09, 0.03
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7827.6 total, 6684.6 free, 488.5 used, 805.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7339.2 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   21740 13040 9624 S   0.0   0.2   0:01.91 systemd
    2 root        20   0   3180   2200 2072 S   0.0   0.0   0:00.03 init-syste+
    7 root        20   0   3196   2132 2008 S   0.0   0.0   0:00.00 init
   59 root       19  -1  34056 12560 11496 S   0.0   0.2   0:00.38 systemd-jo+
  107 root       20   0  25264  6552  5128 S   0.0   0.1   0:00.42 systemd-ud+
  118 systemd+   20   0  21344 13232 11108 S   0.0   0.2   0:00.29 systemd-re+
  119 systemd+   20   0  91036  7996  7024 S   0.0   0.1   0:00.19 systemd-ti+
  174 root       20   0   4244   2680  2432 S   0.0   0.0   0:00.01 cron
  175 message+   20   0   9600   5360  4664 S   0.0   0.1   0:00.19 dbus-daemon
  195 root       20   0  17984  8888  7848 S   0.0   0.1   0:00.22 systemd-lo+
  204 root       20   0 1756104 12996 10888 S   0.0   0.2   0:00.32 wsl-pro-se+
  228 root       20   0   3124   1976  1836 S   0.0   0.0   0:00.01 agetty
  232 syslog     20   0  222516  6036  4664 S   0.0   0.1   0:00.20 rsyslogd
  282 root       20   0 107032 23136 13680 S   0.0   0.3   0:00.43 unattended+
  367 root       20   0   6672   4496  3728 S   0.0   0.1   0:00.02 login
  423 root       20   0  20320 11472  9352 S   0.0   0.1   0:00.30 systemd
  424 root       20   0   21168  3504  1792 S   0.0   0.0   0:00.00 (sd-pam)
  450 root       20   0   6080   5324  3676 S   0.0   0.1   0:00.05 bash
  587 root       20   0   3184   1112   980 S   0.0   0.0   0:00.00 SessionLea+
  588 root       20   0   3200   1252  1108 S   0.0   0.0   0:00.06 ReLay(592)
  592 root       20   0   6080   5368  3668 S   0.0   0.1   0:00.09 bash
  633 root       20   0   9308   5684  3488 R   0.0   0.1   0:00.01 top

```

Overall Observation:

The practical successfully demonstrated important Linux operating system concepts, including process creation, command execution, process synchronization, hardware-resource identification,

and process monitoring. The use of `fork()`, `exec()`, and `wait()` demonstrated how Linux creates and manages processes, while commands such as `uname`, `lscpu`, `lsblk`, `ps`, and `top` provided information about system hardware and running processes. The practical helped demonstrate how the operating system manages hardware resources and provides controlled services to user programs.